

Optimal Investments in Volatility

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January 2008

Abstract

Volatility has evolved as an attractive new asset class of its own. The most common instruments for trading volatility are variance swaps. Mean returns of DAX and ESX variance swaps over the time period from 1995 to 2004 are strongly negative, and only part of the negative premium can be explained by the negative correlation of variance swap returns with stock market indices. We analyze the implications of this observation for optimal portfolio compositions. Mean-variance efficient portfolios are characterized by sizable short positions in variance swaps. Typically, the stock index is also sold short to achieve a better portfolio diversification. To be able to capture heterogeneous preferences for higher moments, we use a variant of the Polynomial Goal Programming method. We assume that investors strive for a high Sharpe ratio, high skewness and low kurtosis. Our analysis reveals that a balanced trade-off between Sharpe ratio and skewness often does not exist. Investors are advised to hold the extreme portfolios (Sharpe ratio driven, skewness driven or kurtosis driven) and avoid to be stuck in the middle. This ‘all-or-nothing’ characteristic is reflected in jumps of asset weights when certain thresholds of preference parameters are crossed. These empirical findings can explain why many investors are so reluctant to implement option-based short-selling strategies.

JEL classification: G10; G12; G13

Keywords: Variance Swap; Volatility Risk Premium; Portfolio Analysis; Higher Moments; Polynomial Goal Programming; Hedge Funds

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1 Introduction

Apart from its prominent role as financial risk measure, volatility has evolved to become an established asset class of its own. Volatility can be traded by creating contracts whose payoff depends on the realized volatility during the time to maturity. The buyer of the contract is 'long volatility' and receives a payment from the seller when realized volatility is high. The seller, who is 'short volatility', benefits from a low realized volatility. There are various volatility instruments which differ in the details of the payoff function.

Initially, the main motive for trading volatility was to manage the risk exposure in option positions and to control the vega exposure independently of the position's delta and gamma. With the growth of the volatility trading segment, other market participants became aware of volatility as a new investment vehicle. Investors are usually attracted by the strong negative correlation between volatility movements and stock index returns. One common explanation for this phenomenon is the so-called leverage effect. It describes the fact that the risk of the equity position increases when stock prices fall and the market value of debt stays constant. The opposite relationship holds when stock prices rise, thus decreasing the leverage ratio in market value terms.¹ According to this explanation, the negative correlation is stock return driven. In contrast, the competing feedback hypothesis assumes that volatility changes are the reason for this effect: higher volatility induces a higher risk premium and therefore higher discount rates, thus lowering fundamental stock values. Another explanation why volatility and stock prices should be negatively correlated is to prevent arbitrage opportunities in options markets in the presence of a volatility skew (see Hafner (2004), p. 46). From an investor's point of view, it seems attractive that the negative correlation between volatility and stock index returns is particularly pronounced in stock market downturns, thus offering protection against stock market losses when it is needed most. This is one reason why financial advisors often recommend the 90/10 rule of investing 10% of a stock portfolio in volatility.

Empirical studies, however, indicate that this kind of downside or crash protection might be expensive. The volatility risk premium is found to be strongly negative in the US as well as in the European stock markets (see, e.g., Jackwerth and Rubinstein (1996), Chernov and Ghysels (2000), Coval and Shumway (2001), Pan (2002), Bakshi and Kapadia (2003), Eraker et al. (2003), Driessen and Maenhout (2003), Doran and Ronn (2004a), Doran and Ronn (2004b), Bondarenko (2004), Moise (2004), Santa-Clara and Yan (2004), Carr and Wu (2005), Hafner and Wallmeier (2007)). This means that long volatility positions provide expected returns below the risk-free interest rate. There is some evidence that the magnitude of the premium depends on the level of implied volatility and the time to maturity (see Bliss and Panigirtzoglou (2004)), the buying pressure for index puts (see Bollen and Whaley (2004)) and the uncertainty of forward volatility (see Carr and Wu (2005)). The estimated premium has significant temporal dependencies (see Bollerslev et al. (2005) and Santa-Clara and Yan (2004)). All in all, the negative premium appears to be too large to be explained by the negative covariance with

¹ This relationship between stock returns and leverage corresponds to an argument by Welsh (2004) who proposes that changes in leverage are only due to changes in the market value of equity. See also Bessler et al. (2008) for European evidence. However, according to empirical findings of Figlewski and Wang (2000), the leverage changes cannot fully explain the extent of the negative correlation of stock return and volatility changes.

market returns within standard (partial) equilibrium models (see Carr and Wu (2005), Hafner and Wallmeier (2007)). Consequently, selling volatility on a regular basis might be a profitable strategy. This is compatible with the finding of Bondarenko (2004) that the variance risk factor accounts for a considerable portion of historical hedge fund returns.

It is interesting to note that a strategy of systematically selling volatility earns, on average, a positive risk premium. The economic rationale behind this empirical observation is that investors receive a premium for providing other market participants with insurance against potential market losses. However, the return distribution of short volatility is negatively skewed (see Carr and Wu (2005) and Hafner and Wallmeier (2007)), which means that the strategy will cause substantial losses if the volatility suddenly rises. Historically this has happened in situations of (financial) crisis. A prominent example is the collapse of the Long Term Capital Management (LTCM) hedge fund. LTCM had built up large short positions in equity volatility in the forefront of the Asian and Russian crisis in 1997/98. When volatility dramatically increased, LTCM was confronted with huge losses. The losses from short volatility trades were estimated to be about 1.3 bn USD – roughly 30% of LTCM's total loss (see Lowenstein (2000)).

The negative skewness in the return distribution of a (short) volatility trading strategy makes the volatility asset class different from traditional investment vehicles, such as stocks and bonds. This has a number of important implications for asset allocation and risk management. First, higher moments such as skewness and kurtosis have to be considered in the asset allocation decision process in addition to the first two moments of the portfolio return distributions. Second, in managing the risk of a short variance strategy, risk measures should be used that are able to quantify the financial impact of potentially damaging events. One such measure is the conditional value at risk (CVaR). Third, losses of a short variance strategy are potentially unlimited.² One way to alleviate the effects of an adverse market movement is to hedge part of the tail risk of the strategy, e.g. by buying out-of-the-money variance call options. These options are now regularly traded in the OTC market.

Although there are many studies on the variance risk premium, so far the implications for investors have rarely been addressed in any detail. In order to draw any conclusions from the negative risk premium from an investor's point of view, it is necessary to specifically analyze optimal portfolio compositions in realistic settings. This is exactly the focus of our research. We are particularly interested in the trade-off between Sharpe ratio, skewness and kurtosis. This trade-off is not only important for the short volatility strategy presented here, but for many other hedge fund strategies that produce heavily skewed return distributions.³ In general, such strategies are characterized by small profits in 'normal' situations and potentially large losses when some rare event occurs.

The results of our analysis suggest that a balanced trade-off between Sharpe ratio, skewness and kurtosis often does not exist. Investors are advised to hold the extreme portfolios – Sharpe ratio driven, skewness driven or kurtosis driven – and avoid to be stuck in the middle. This 'all-or-nothing' characteristic is reflected in jumps of asset weights when certain thresholds of

² In practice, variance contracts often exhibit a cap of 2.5 times the current swap rate.

³ In analysing hedge fund returns, Bondarenko (2004) finds a significant relation to variance returns. This indicates that short variance strategies are in part responsible for skewness and kurtosis in hedge fund returns.

preference parameters are crossed. These empirical findings can explain why many investors are so reluctant to implement option-based short-selling strategies.

The rest of the paper is organized as follows. In the next section, we introduce variance swaps as an investment instrument. We synthetically derive values of variance swaps over the time period from 1995 to 2004 using tick-by-tick data for options on the German stock index DAX and the European index Euro STOXX 50 (ESX). The historical variance swap returns serve as a basis for our portfolio analysis (Section 3). The mean-variance framework (Subsection 3.1) neglects the return properties of negative skewness and excess kurtosis. These are taken into account implicitly in expected utility maximization (Subsection 3.2) and explicitly in the *Polynomial Goal Programming (PGP)* optimization (Subsection 3.3). The paper concludes with a brief summary.

2 Variance Swap Returns

2.1 Variance Swaps and Valuation

Variance swaps, which are also called variance contracts, offer pure exposure to the realized future variance. The payment at expiration is equal to the difference between the annualized variance of log stock returns and the swap rate fixed at the outset of the contract. If this difference is positive, the swap buyer receives payment from the swap seller. If it is negative, the swap seller receives payment from the buyer. The swap rate is chosen such that the contract has zero present value. Thus, the swap rate can be interpreted as the risk neutral expectation of the unconditional future variance. Variance swaps on the most common stock indices now have an active over-the-counter market. The market growth was possible because of the theoretical research deriving a robust replication strategy. This strategy consists of a continuously adjusted stock holding and a static options portfolio including long positions in out-of-the-money (OTM) options for all strikes from zero to infinity (see Neuberger (1994)). Based on this replication, we can express the fair swap delivery price K_{VARS} as:

$$K_{VARS} = \frac{2}{T}e^{rT} \left(\int_0^{F_0(T)} \frac{1}{K^2} P_0(K, T) dK + \int_{F_0(T)}^{\infty} \frac{1}{K^2} C_0(K, T) dK \right), \quad (1)$$

where $C_0(K, T)$ and $P_0(K, T)$ denote the current market price of a put and a call option of strike K and maturity T , r is the risk-free rate and $F_0(T)$ is the stock's T -maturity forward price. We assume that there is no credit risk involved. According to formula (1), the replication portfolio includes out-of-the-money put and call options with weights inversely proportional to the square of their strike. This weighting scheme is necessary to obtain a constant portfolio exposure with respect to realized variance independent of stock price moves (see Demeterfi et al. (1999) for a detailed derivation and economic motivation).

In a perfect market, the replication is exact if options with arbitrary strikes are available and the stock price process is continuous. If stock price jumps occur and the number of strike prices is limited, the formula still offers a good approximation in realistic settings (see Carr and Wu (2005)). Our comparison with OTC quotes from two major investment banks in November and

December 2004 confirms that the theoretical values from (1) are close to market prices. With the exception of two days, the theoretical values always fall into the spread between bid and ask quotes. This spread seems sufficiently narrow (about 1 volatility point) to make the comparison meaningful.

To apply formula (1) in the empirical analysis, we first estimate the strike price structure of implied volatilities ('smile'). The strike-dependent implied volatilities are then put into the Black-Scholes formula (or equivalently the Black (1976) model, if forward prices for the underlying asset are used) to obtain the strike price structure of option prices $C_0(K, T)$ and $P_0(K, T)$.

2.2 Data and Estimation

Our database contains all reported transactions of the two stock index options and futures with the highest trading volume in Europe. Option and future data for the German stock index DAX are available from January 1995 to December 2004, those on the Euro STOXX 50 index (ESX) from January 2000 to December 2004. Both European style options ('ODAX' and 'OESX') are traded on the joint German and Swiss options and futures exchange Eurex.⁴

To calculate an implied volatility for each transaction, it is crucial to accurately match the corresponding forward price. As we use time-stamped tick-by-tick data, matching of option and future prices is straightforward. We apply the method of Hafner and Wallmeier (2001) to account for dividend effects and to ensure put-call-parity consistent estimates of implied volatilities.

For each trading day and each time to maturity available on that day, we estimate a smooth curve of implied volatilities across strike prices. Let K denote the strike price of an option with time to maturity $T - t$. Each trade is assigned a moneyness according to:

$$M(t, T, F_t(T), K) = \frac{\ln\left(\frac{K}{F_t(T)}\right)}{\sqrt{T - t}},$$

where $F_t(T)$ is the (intraday) forward price at the time of this trade.

Suppressing the arguments of moneyness, we chose the cubic regression function:

$$\sigma = \beta_0 + \beta_1 M + \beta_2 M^2 + \beta_3 D \cdot M^3 + \varepsilon, \quad (2)$$

where σ is the implied volatility, β_i , $i = 0, 1, 2, 3$ are regression coefficients, ε is a random error, and D is a dummy variable which assumes the value of one for positive moneyness and of zero otherwise. The dummy variable accounts for the asymmetry pattern of implied volatilities around the at-the-money strike ($M = 0$). Typically, the 'smile' is better characterized by a 'skew', with the negative relation between implied volatility and moneyness extending clearly beyond $M = 0$. Only when the call (put) is deep out-of-the-money (in-the-money) the implied volatility function forms a minimum and eventually rises slightly. A quadratic or cubic regression without differentiating between $M \leq 0$ and $M > 0$ does not capture this increase. The regression function (2) is twice differentiable, which ensures that the corresponding risk-neutral density is continuous.

⁴ We are very grateful to the Eurex for providing the data.

The smile estimation according to Eq. (2) is based on all trades on one day in options with the same time to maturity. In order to obtain an estimate of the smile for a given, pre-specified time to maturity of x calendar days, we linearly interpolate between the implied variances of the two neighbouring maturities which are available (see, e.g., Wilmott (1998), p. 290). In this study, we choose $\tau = 45$, because ODAX and OESX option series with lifetimes between 30 and 60 days are the most liquid contracts, which ensures an accurate estimation of the smile. The average R^2 -coefficient over all trading days of one year is 92% in 1995 and larger than 95% in all later years.

We assume the smile function of Eq. (2) to be valid in a moneyness-range between the lowest and highest moneyness of all observations. Outside this range, following Carr and Wu (2005) and Jiang and Tian (2005), we assume that implied volatilities are constant at the volatility level of the relevant moneyness boundary. This corresponds to a conservative estimate of the fair values of options far in and out-of-the-money. Other extrapolation techniques would provide higher variance swap rates and (even) lower variance returns.

Having estimated the smile structure, we convert implied volatilities from the regression function (2) into call and put option prices (functions $C_0(K, T)$ and $P_0(K, T)$). It is then straightforward to numerically calculate the integrals in (1). The calculations result in an estimate of the variance swap value for each trading day in the sample period.

2.3 Return Distributions

Institutional investors will typically hold a variance swap position over a certain horizon. We therefore assume a buy-and-hold strategy over the lifetime of each contract (45 days). In order to obtain a complete picture of possible returns over this holding period, we assume that one contract is bought on each trading day during the period under study. The resulting return series is strongly autocorrelated because of overlapping return periods (day 1 to 45, day 2 to 46, etc.). We account for this problem by using serial-dependence adjusted Newey/West (1987)-standard errors (with a lag of 33 trading days, corresponding to 45 calendar days). As a robustness check, we repeated all tests for non-overlapping periods and obtained similar standard error estimates.

Table 1 presents summary statistics of the distribution of DAX and ESX index log returns (r_S) and variance swap log returns (r_{VARS}). The mean variance swap returns are significantly negative in all time periods.⁵ The implication of this observation is that the variance swap rates over-estimate the subsequently realized variance. Indeed, it is well-known that ATM implied volatility tends to be higher than subsequently realized volatility. Due to the option's smile, variance swap rates are even higher than the ATM variance, which increases the spread further. As presented in Table 1, it was more profitable to initiate a short position in DAX variance swaps in the second period from 2000 to 2004 than in the first period from 1995 to 1999. In the second period, however, it would have been even more advantageous to shorten ESX variance swaps. The mean discrete return for 45-days ESX variance swaps in the period from 2000 to 2004 is minus 19.3%, while it is only minus 13.6% for DAX variance swaps. The above results indicate

⁵ There is only one mean return (DAX, period 2000-2004) that is not statistically significant at the 5% level.

		DAX			ESX
		95-04	95-99	00-04	00-04
r_S	Mean	0.008	0.022	-0.013	0.012
	Median	0.022	0.029	<-0.001	-0.006
	Minimum	-0.437	-0.337	-0.437	-0.340
	Maximum	0.249	0.245	0.249	0.221
	Std. dev.	0.089	0.075	0.102	0.078
	Skewness	-0.902	-0.848	-0.670	-0.554
	Kurtosis	1.898	2.219	0.992	0.964
r_{VARS}	Mean	-0.271	-0.235	-0.307	-0.360
	Median	-0.324	-0.259	-0.387	-0.419
	Minimum	-1.569	-1.569	-1.547	-1.658
	Maximum	1.586	1.087	1.586	1.295
	Std. dev.	0.498	0.472	0.521	0.510
	Skewness	0.560	0.176	0.890	0.676
	Kurtosis	3.690	2.781	1.475	3.545
$Corr$		-0.476	-0.421	-0.574	-0.534

Table 1: Summary Statistics for discrete returns and log returns of variance swaps on the DAX and ESX index with a time to maturity 45 calendar days. Robust t -statistics are calculated using the Newey-West estimator for the standard deviation with a lag of 33.

that, on average, investors are willing to accept a heavily negative risk premium for being long in realized variance. Equivalently, investors who are providing insurance to the market, i.e., who are sellers of variance, require a significantly positive risk premium.

The sample Pearson correlation coefficient of r_S and r_{VARS} is given as $Corr$ in Table 1. As expected, it is significantly negative in all cases.

The payoff and discrete return distributions of DAX and ESX variance swaps are clearly non-normal: they show positive skewness and excess kurtosis. The log transformation of discrete returns to continuously compounded returns, however, reduces the skewness estimate for DAX (ESX) variance swap returns in the period from 1995-2004 (2000-2004) from 2.689 (2.282) to 0.560 (0.676) and the kurtosis estimate from 13.246 (8.793) to 3.690 (3.545). The log return distributions appear to be close to normal, although standard tests (Jarque-Bera test, Kolmogorov-Smirnov goodness-of-fit test, etc.) reject normality. Figure 1 illustrates these findings. It shows the empirical density functions for the log returns of DAX and ESX variance swaps along with normal distributions having the same means and the same variances as those estimated from the samples. The distributions of DAX and ESX log variance swap returns are in most cases close to each other (see Figure 2). In some cases, however, they strongly deviate. For example, during the period July 30th to September 10th, 2001 the log return of DAX variance swaps was significantly higher than the log return of ESX variance swaps (points marked as crosses in the graph). Investors who bought variance swaps in this period realized substantial profits, since September 11 is within the swap's time to maturity, so that realized variance was extremely

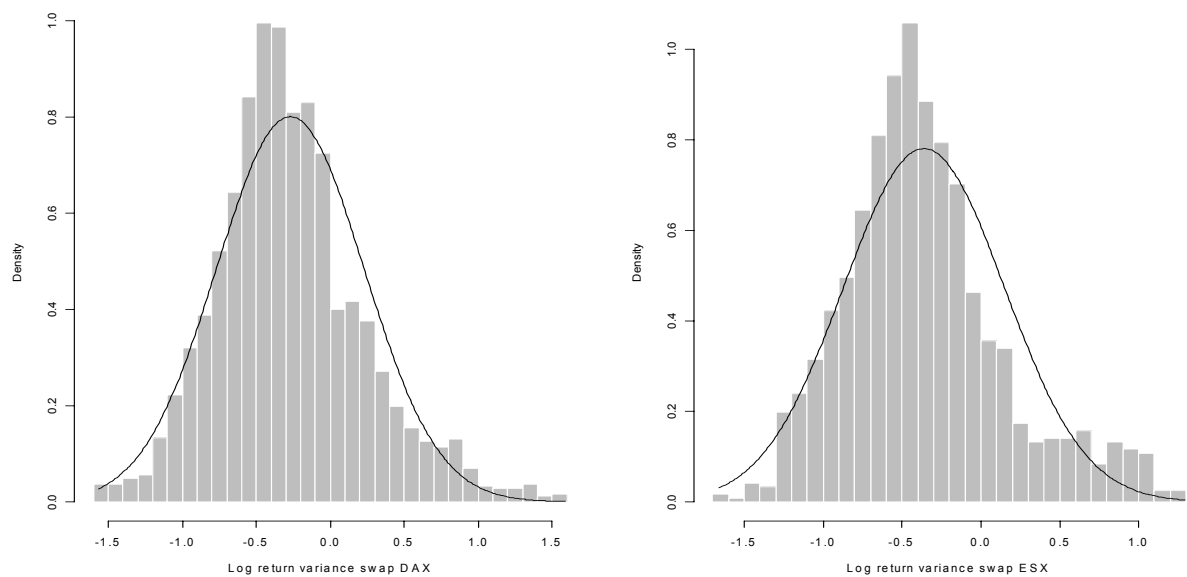


Figure 1: Histogram of 45 calendar day log returns of DAX variance swaps (left graph) and ESX variance swaps (right graph) over the sample periods 1995-2004 (DAX) and 2000-2004 (ESX).

high.⁶

⁶ See Hafner and Wallmeier (2007) for a description of variance swap rates during other crises.

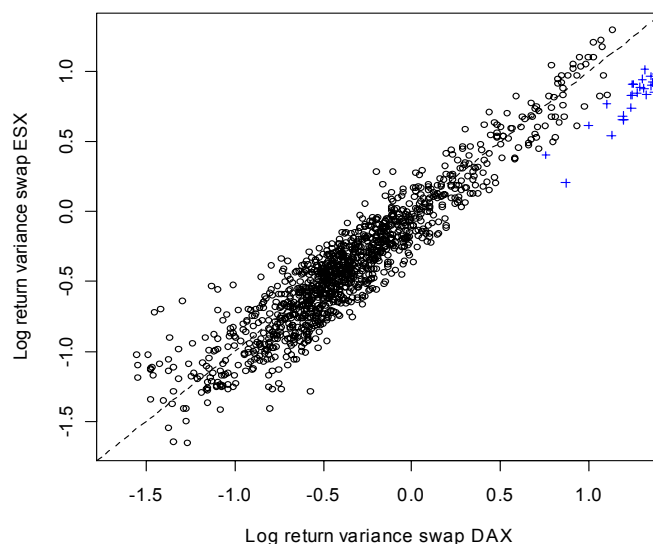


Figure 2: Log return variance swap DAX versus log return variance swap ESX within the period 2000-2004. The crosses mark all dates where September 11 is within the swap's time to maturity.

3 Portfolio Analysis

3.1 Mean-Variance Efficient Portfolios

Given the historical return distributions, the interesting question from an investor's perspective is how they should structure their portfolios in order to profit from the attractive return opportunities that are offered either by selling variance swaps or by the diversification benefit that buying positions provide. To tackle this question, we first compare mean-variance efficient portfolios with and without variance swaps and secondly we analyze optimal portfolios under power utility.⁷ In the third subsection, we will explicitly include the skewness and kurtosis preferences of the investor in the analysis by using the *Polynomial Goal Programming* method. Our asset universe always consists of variance swaps, its underlying stock index, and the risk-free asset.

In section 2, we have defined variance swap returns as the ratio of realized variance and the present value of the delivery price K_{VARS} . We thereby assume that the buyer of the swap makes an up-front payment of $e^{-rT}K_{VARS}$ in order to receive a payment of 1 euro times realized variance at delivery. In reality, though, except for the margin requirements, there is no cost to enter into the variance swap contract. Since there is no initial investment, we can neither characterize the profit or loss in relative terms nor can we determine the weight of variance swaps in the investor's portfolio. To overcome this problem, we 'deleverage' the contract by introducing an up-front payment in the form of a risk-free investment. The proceeds of the

⁷ These analyses are based on discrete returns as required by the usual calculations in portfolio theory.

risk-free asset increase the net payoff at maturity. An investment of $e^{-rT}K_{VARS}$ seems to be the natural choice because this amount corresponds to the present value of a net payoff equal to the realized variance. However, this still implies a high degree of leverage compared to investing in stocks, as is evident from the higher return fluctuations. Therefore, as an alternative, we assume a risk-free investment of $f \cdot e^{-rT}K_{VARS}$, where the factor f is chosen such that the volatility of variance swap returns is equal to the index return volatility in our sample period. This makes it easier to interpret the portfolio weights of variance swaps and to compare them to the weights of stocks. It is evident, that the set of mean variance efficient portfolios will be the same for any value of f as long as the risk-free asset is part of the asset universe. We refer to the first return definition ($f = 1$) as *HL* (High Leverage) and to the second definition (same volatility as stock index) as *LL* (Low Leverage).

Figure 3 illustrates the mean-variance analysis for the *LL*-case using estimates from our sample period. The asset universe consists of the DAX index (weight x_S), the DAX variance swap (x_{VARS}) and the risk-free asset (x_{rf}). The sample average and the sample standard deviation of DAX returns over all intervals of 45 days in the period from 1995 to 2004 were 0.76% and 8.66%, respectively. Over the same set of intervals, an *LL*-mean return of -2.05% was observed for DAX variance swaps. Line (1) is the efficient frontier without considering variance swaps ($x_S + x_{rf} = 1$), whereas line (2) represents all combinations of DAX and variance swaps without the risk-free asset ($x_{VARS} + x_S = 1$). If we allow all three assets to enter into the portfolio, we obtain the new efficient line (5). All portfolios on this line are characterized by the same ratio x_{VARS}/x_S but with different weights of the risk-free asset. Pre-specifying the weight x_{rf} and maximizing the Sharpe ratio, we obtain one point on the efficient line (5). For instance, line (4) with tangency portfolio T_1 represents all portfolios with $x_{rf} = 1.1$, line (3) with tangency portfolio T_2 represents all portfolios with $x_{rf} = 2$. As we move on the efficient line towards combinations of higher risk and return, the short sales of the risky part of the portfolio increase, meaning that the sum $x_{VARS} + x_S$ becomes more negative. To the same extent, the weight of the risk-free asset increases. This increase results in a riskier portfolio because the risk-free investment is financed by short selling risky assets. In the *HL*-case of our return definition, we obtain the same efficient line (5). The efficient portfolios are merely characterized by a different ratio of variance swap and stock index weights.

Table 2 summarizes characteristics of mean-variance efficient portfolios, where we restrict portfolio weights to a range between -3.0 and 3.0 .⁸ The first part of the table ('Base case') is based on sample estimates of mean returns, standard deviations and the correlation coefficient. In order to examine the sensitivity of the results to errors in the estimated variance risk premium, we then either increase ('Case 2') or decrease ('Case 3') the mean return of variance swaps by twice the Newey-West standard error of the mean estimate. Leaving all other input parameters constant, we obtain the results shown in the second and third part of Table 2. SR_0 denotes the Sharpe ratio without variance swaps, and SR is the Sharpe ratio of efficient portfolios including variance swaps. It is important to note that our allocation analysis does not take general equi-

⁸ It is reasonable to restrict the magnitude of individual asset positions and the extent to which investors are allowed to sell assets short. Most institutional investors have to meet such restrictions. We assume that the individual long and short positions cannot be larger than three times as much as net investment. However, our results are robust with respect to the values of the upper and lower boundaries.

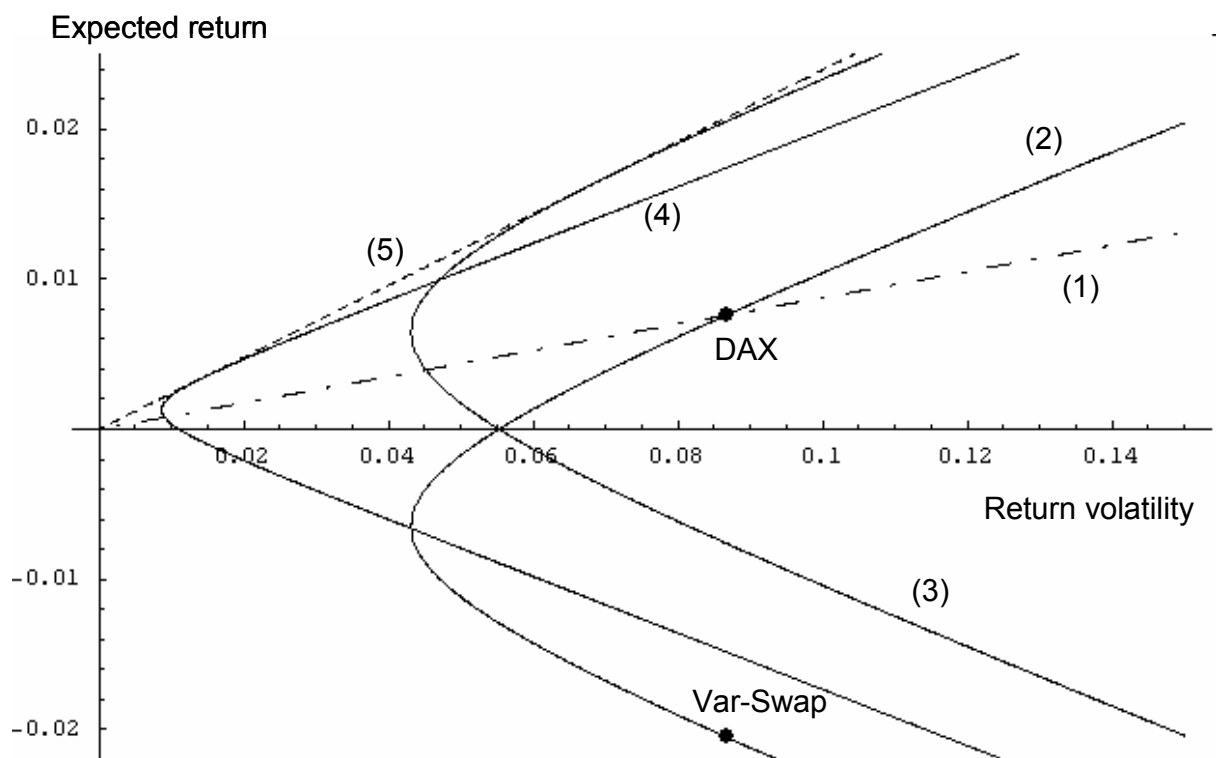


Figure 3: Mean variance analysis of DAX and variance swap investments. The expected (excess) returns and variances are estimated from the sample of return observations in all intervals of 45 days in the period from 1995 to 2004.

		Case 1: Base case					
SR_0		SR	x_{VARS}	x_S	x_{rf}	$LL : \frac{x_{VARS}}{x_S}$	$HL : \frac{x_{VARS}}{x_S}$
DAX 95-04	0.0876	0.2394	< 0	< 0	> 1	6.1140	0.9655
DAX 95-99	0.3632	0.3743	< 0	> 0	< 1	-0.3394	-0.0566
DAX 00-04	-0.1336	0.3728	< 0	< 0	> 1	1.1667	0.1709
ESX 00-04	-0.1740	0.5924	< 0	< 0	> 1	1.2438	0.1816
		Case 2: Higher return of variance swap (+2 STD)					
SR_0		SR	x_{VARS}	x_S	x_{rf}	$LL : \frac{x_{VARS}}{x_S}$	$HL : \frac{x_{VARS}}{x_S}$
DAX 95-04	0.0876	0.0885	< 0	> 0	< 1	-0.1888	-0.0298
DAX 95-99	0.3632	0.4143	> 0	> 0	< 1	0.4814	0.0803
DAX 00-04	-0.1336	0.1399	< 0	< 0	> 1	0.3055	0.0448
ESX 00-04	-0.1740	0.3059	< 0	< 0	> 1	0.8882	0.1297
		Case 3: Lower return of variance swap (-2 STD)					
SR_0		SR	x_{VARS}	x_S	x_{rf}	$LL : \frac{x_{VARS}}{x_S}$	$HL : \frac{x_{VARS}}{x_S}$
DAX 95-04	0.0876	0.4413	< 0	< 0	> 1	3.0466	0.4811
DAX 95-99	0.3632	0.5260	< 0	> 0	< 1	-3.1768	-0.5301
DAX 00-04	-0.1336	0.6683	< 0	< 0	> 1	1.4188	0.2079
ESX 00-04	-0.1740	0.8980	< 0	< 0	> 1	1.4043	0.2050

Table 2: Characteristics of mean-variance efficient portfolios.

librium conditions into account. Because variance swaps are on aggregate in zero net supply, large positive or negative weights of these instruments are generally not compatible with general equilibrium.

The weight of variance swaps is negative with the exception of the first subperiod for DAX swap returns in Case 2. Typically, the stock index also enters into the efficient portfolios with a negative weight. This short-selling of the stock index fits to the short position in variance swaps, because in this way, investors benefit from the negative correlation so that they can achieve a better diversification. In Case 2, however, selling the index short is only suitable in the second subperiod. The Sharpe ratio SR for DAX is 0.37 in both subperiods, but only 0.24 in the full sample. This is due to the different portfolio structures in the two subperiods. In fact, a long index position is required in the first period and a short position in the second period to reach the higher Sharpe ratio of 0.37. If we construct only one portfolio for the entire sample period, this would be suboptimal in both subperiods.

The optimal ratio of x_{VARS} and x_S strongly differs along time period and underlying index. In the Cases 1 and 3, the variance swap weight typically exceeds the stock index weight. It is interesting to note that the short position in variance swaps is not necessarily extended when assuming more strongly negative variance swap returns. For instance, over the full period of DAX returns, efficient portfolios are characterized by a ratio x_{VARS}/x_S of 6.1, compared to 3.0 in Case 2. Thus, variance swaps are less aggressively sold compared to the underlying index, although the absolute variance risk premium has been raised. The reason for this counter-intuitive observation is that the risk reduction resulting from less divergent weights x_{VARS} and x_S is larger than the loss in expected return.

3.2 Backtesting under Power Utility

In Table 3, we present optimal portfolio weights for an investor who maximizes his expected utility based on the power utility function with risk aversion parameter α . As in the previous section, we differentiate between three cases. In the base case, the bivariate distribution of excess returns of variance swaps and the underlying stock index is set to be equal to the observed distribution in the sample period. The columns ‘+2 STD’ result from shifting all variance swap returns by twice the Newey-West adjusted standard error of the volatility risk premium. In the case ‘-2 STD’, the adjustment shifts into the opposite direction so that the negative risk premium gets even larger. The table is based on variance swaps that are levered so that their sample return volatility equals the volatility of the stock index return (LL -definition of previous section). As before, the portfolio weights are restricted to lower and upper bounds of -3.0 and 3.0 , respectively.

The results can be summarized as follows: In the base case, the weights x_{VARS} are all negative. The size of the short position decreases with a higher degree of risk aversion. The weight of the variance swap is always lowest in the case ‘-2 STD’ and highest in the case ‘+2 STD’. During the second subperiod, the investor also takes a short position in stocks, but its weight is smaller than x_{VARS} . Since the optimal portfolio typically contains short positions in the index and in variance swaps, the risk-free asset often has a large weight. Our analysis of the period from 1995 to 1999

provides substantially different results. The weights of the short position in variance swaps are rather small, and the risky part of the portfolio is strongly concentrated in long index holdings. This is certainly due to the high stock returns during that period of ‘irrational exuberance’. Overall, the results are similar to the preceding mean variance analysis.

DAX 1995 - 2004									
α	Base case			+2 STD			-2 STD		
	x_{VARS}	x_S	x_{rf}	x_{VARS}	x_S	x_{rf}	x_{VARS}	x_S	x_{rf}
1	-1.56	-0.31	2.87	-0.10	0.89	0.21	-1.73	-0.27	3.00
1.5	-1.22	-0.25	2.47	-0.08	0.60	0.48	-1.58	-0.42	3.00
2	-0.98	-0.20	2.18	-0.06	0.45	0.61	-1.44	-0.56	3.00
5	-0.44	-0.09	1.53	-0.03	0.18	0.85	-0.73	-0.33	2.06
10	-0.22	-0.04	1.26	-0.01	0.09	0.92	-0.39	-0.17	1.56
DAX 1995 - 1999									
α	Base case			+2 STD			-2 STD		
	x_{VARS}	x_S	x_{rf}	x_{VARS}	x_S	x_{rf}	x_{VARS}	x_S	x_{rf}
1	-0.21	2.87	-1.66	0.80	3.20	-3.00	-2.65	0.65	3.00
1.5	-0.23	2.17	-0.94	1.13	2.87	-3.00	-2.33	0.41	2.92
2	-0.21	1.71	-0.50	1.34	2.66	-3.00	-1.88	0.37	2.51
5	-0.12	0.73	0.39	0.73	1.30	-1.03	-0.84	0.19	1.65
10	-0.06	0.37	0.69	0.37	0.66	-0.03	-0.43	0.10	1.33
DAX 2000 - 2004									
α	Base case			+2 STD			-2 STD		
	x_{VARS}	x_S	x_{rf}	x_{VARS}	x_S	x_{rf}	x_{VARS}	x_S	x_{rf}
1	-1.19	-0.81	3.00	-0.40	-1.60	3.00	-1.64	-0.36	3.00
1.5	-1.10	-0.90	3.00	-0.30	-1.10	2.40	-1.43	-0.57	3.00
2	-1.05	-0.95	3.00	-0.23	-0.83	2.06	-1.30	-0.70	3.00
5	-0.67	-0.69	2.36	-0.10	-0.34	1.44	-1.01	-0.98	2.99
10	-0.34	-0.34	1.68	-0.05	-0.17	1.22	-0.53	-0.50	2.03
ESX 2000 - 2004									
α	Base case			+2 STD			-2 STD		
	x_{VARS}	x_S	x_{rf}	x_{VARS}	x_S	x_{rf}	x_{VARS}	x_S	x_{rf}
1	-1.64	-0.36	3.00	-0.81	-1.19	3.00	-2.18	0.18	3.00
1.5	-1.42	-0.58	3.00	-0.87	-1.13	3.00	-1.85	-0.15	3.00
2	-1.31	-0.69	3.00	-0.89	-1.11	3.00	-1.64	-0.36	3.00
5	-1.08	-0.92	3.00	-0.73	-0.85	2.58	-1.20	-0.80	3.00
10	-0.70	-0.65	2.35	-0.37	-0.43	1.80	-0.99	-0.88	2.87

Table 3: Optimal portfolio weights under power utility.

3.3 Polynomial Goal Programming

Maximization of expected utility is based on the return distribution as a whole and therefore implicitly takes into account the higher moments such as skewness and kurtosis. However, it is generally not evident how to adjust the risk aversion parameter in the utility function to capture heterogeneous investor preferences with respect to the different moments of the return distribution. To be able to explicitly control the influence of higher moments, we use *Polynomial Goal Programming (PGP)* which was introduced by Lai (1991) and later applied by Chunnachinda et al. (1997), Prakash et al. (2003), Sun and Yuxing (2003), Canela and Colla (2004), and Davies et al. (2004). This method allows us to incorporate different objectives whose relative importance depends on individual preference parameters. We assume that investors aim at a high Sharpe ratio SR , a high (positive) skewness SK and a low kurtosis KT , where

$$SR = \frac{E[R]}{\sigma}; \quad SK = \frac{E[(R - E[R])^3]}{\sigma^3}; \quad KT = \frac{E[(R - E[R])^4]}{\sigma^4}.$$

R denotes excess returns and σ is the standard deviation of R . If the investor attempts to find an optimal trade-off between these objectives, she has to balance a gain in one objective against a loss in another. We assume that her overall assessment is based on the discrepancy between the degree of actual achievement and the maximal degree of achievement ($SR^{\max}$, $SK^{\max}$, $KT^{\min}$) that is possible if one objective alone is considered. Since Sharpe ratio, skewness and kurtosis are of different magnitudes, we have to rescale these measures to make them comparable. Therefore, we express the discrepancies between the maximal degree and the actual degree of achievement in *relative* terms. Our overall objective function to be minimized is defined as:

$$Z(x_{VARS}, x_S) = (1 + d_{SR}(x_{VARS}, x_S))^{\alpha} + (1 + d_{SK}(x_{VARS}, x_S))^{\beta} + (1 + d_{KT}(x_{VARS}, x_S))^{\gamma}, \quad (3)$$

where:

$$d_{SR}(x_{VARS}, x_S) = \frac{SR^{\max} - SR(x_{VARS}, x_S)}{SR^{\max}}, \quad (4)$$

$$d_{SK}(x_{VARS}, x_S) = \frac{SK^{\max} - SK(x_{VARS}, x_S)}{SK^{\max}}, \quad (5)$$

$$d_{KT}(x_{VARS}, x_S) = \frac{KT(x_{VARS}, x_S) - KT^{\min}}{KT^{\min}}. \quad (6)$$

The parameters $\alpha \geq 0$, $\beta \geq 0$, and $\gamma \geq 0$ express the investor's preference with respect to the Sharpe ratio, skewness and kurtosis.⁹ The weight of the risk-free asset is given by $x_{rf} = 1 - x_{VARS} - x_S$.

The ratios d_{SR} , d_{SK} , and d_{KT} according to Eq. (4) to (6) are invariant with respect to multiplying x_{VARS} and x_S by the same constant $c > 0$. Such a transformation changes portfolio variance by the factor c^2 . This means that the solution of the PGP optimization problem is compatible with any degree of portfolio variance. The desired level of variance is simply achieved

⁹ Skewness preference is an implication of decreasing absolute risk aversion. The experimental and empirical evidence of preference for positively skewed returns is summarized in Harvey et al. (2004). For a discussion of kurtosis preference, see Dittmar (2002).

by adjusting the weight of the risk-free asset without changing the structure of the risky sub-portfolio. Thus, we can pre-specify the variance without restricting the space of solutions. For convenience, we choose $\sigma^2 = 1$. As before, the expected values needed as input parameters are estimated by the respective mean values of the historical return sample. In addition to the parameters entering in the mean-variance optimization, the parameter set also includes the skewness and kurtosis of variance swap and underlying index returns as well as the co-movements $E[(R_{VARS} - E[R_{VARS}])^x, (R_S - E[R_S])^y]$ with $(x, y) \in \{(1, 2), (2, 1), (1, 3), (3, 1), (2, 2)\}$.

Figure 4 illustrates some important elements of the optimization results for DAX variance swaps over the entire period from 1995 to 2004. Since short selling is allowed, the asset weights for portfolios with the same variance are given by an ellipse that is concentric about the point of the minimum variance portfolio characterized by $x_{VARS} = x_S = 0$ (and therefore $x_{rf} = 1$). The ellipse shown in Figure 4a) represents a portfolio variance of 1. Corresponding to this high volatility of 100%, the asset weights are rather extreme. But as already mentioned, it is only the *ratio* of asset weights that matters in our analysis. Their magnitude can be easily scaled down to be compatible with a lower variance.

On the line in Figure 4a), we highlight the portfolio compositions resulting from separately optimizing one of the objectives, i.e. either Sharpe ratio, skewness or kurtosis.¹⁰ As is already known from the mean-variance analysis in Section 3.1, the portfolio with the highest Sharpe ratio is short in variance swaps as well as in the underlying index, with a much larger (negative) weight on variance swaps. Maximizing portfolio skewness results in a completely different portfolio which is at the opposite side of the isovariance ellipse. The $KT^{\min}$ -portfolios are again very different from both the $SR^{\max}$ - and $SK^{\max}$ -portfolio. This illustrates the extent to which the three objectives are conflicting. The conflict between the skewness and sharpe ratio objectives is also apparent from the different alignment of the isovariance lines of $1 - d_{SR} = SR/SR^{\max}$ and $1 - d_{SK} = SK/SK^{\max}$ against x_{VARS} in Figure 4b). The best portfolio with respect to skewness provides almost the worst results with respect to the Sharpe ratio, and vice versa.

The investor's optimal portfolio depends on her preference structure as represented by the triple (α, β, γ) . Figure 4c) shows how the asset weights x_{VARS} and x_S depend on β when the other parameters are fixed at $\alpha = 1$ and $\gamma = 0$. In the case of $\beta = 0$, the Sharpe ratio is the only relevant objective, so that the asset weights correspond to the $SR^{\max}$ -portfolio. With increasing β , skewness becomes more and more important relative to the Sharpe ratio. At first, the extent of short selling stocks rises without significant changes in the variance swap position. The optimal portfolio thus moves from the point $SR^{\max}$ in Figure 4a) in the direction of the lower $KT^{\min}$ -portfolio. This smooth adjustment stops when β exceeds a level of about 1.32. At this threshold, the portfolio weights jump to the $SK^{\max}$ -portfolio composition. The reason for this abrupt rearrangement of the optimal portfolio is that the objective function $Z(x_{VARS}, x_S)$ has two local minima, one where the positive variance swap weight is large, and the other where the negative weight is large (see Figure 4d). Portfolios with x_{VARS} in-between these values are apparently not attractive. For β lower than 1.32, the investor chooses the 'left' local minimum which represents a structure close to the $SR^{\max}$ -portfolio, and for β higher than 1.32, the investor chooses the 'right' local minimum which is almost identical to the $SK^{\max}$ -portfolio. Thus, the

¹⁰ There are two portfolios with minimal kurtosis, since $KT(x_{VARS}, x_S) = KT(-x_{VARS}, -x_S)$.

trade-off between Sharpe ratio and skewness seems to be such that investors cannot always find a reasonable compromise. The core of the matter is that they primarily face a decision between two real *alternatives* so that the optimization problem resembles an ‘all or nothing’ decision.

To further investigate this observation, Figure 4e) plots the relative degree of achievement of the Sharpe ratio objective (measured by $1 - d_{SR} = SR/SR^{\max}$) against the degree of achievement of the skewness objective (measured by $1 - d_{SK} = SK/SK^{\max}$) for all portfolios with a one unit variance. Given an arbitrary level of skewness achievement, investors will always prefer the portfolio with a higher Sharpe ratio. Therefore, only the upper line is relevant and might be called ‘efficient’. On the extreme left, the Sharpe ratio has a maximum, on the extreme right, skewness has a maximum. The efficient line is concave in its left and convex in its right part. Starting on the left, skewness can at first be improved without much loss in Sharpe ratio. But the higher the slope coefficient (in absolute terms) of the efficient line, the more ‘expensive’ in terms of Sharpe ratio is a further increase in portfolio skewness. If an investor is willing to pay this price, it makes sense to even go a step further and choose a portfolio near $SK^{\max}$. The reason is that the slope eventually diminishes again, rendering a further increase in skewness less costly. With increasing β in the setting ($\alpha = 1, \beta, \gamma = 0$), the optimal portfolio moves from $SR^{\max}$ to T ($\beta = 1.32$) and then jumps to a point near $SK^{\max}$. The middle part of the efficient line is no-man’s-land: Those who go beyond T will not stop before reaching $SK^{\max}$.

If we include kurtosis preference into the analysis, the portfolio structure once again does not shift continuously towards the kurtosis minimal portfolio, but it exhibits a pronounced jump at a certain γ -threshold (see Figure 4e). As is already known, in the case of $\alpha = 1$ and $\beta = 1.33$, the optimal portfolio without considering kurtosis ($\gamma = 0$) is close to the $SK^{\max}$ -portfolio. The asset weights are stable as long as γ is below 0.21. But as soon as γ exceeds this threshold, the optimal solution shifts to one of the two $KT^{\min}$ -portfolios. In the case of DAX variance swaps, the $KT^{\min}$ -portfolio with negative asset weights is chosen since it represents a more attractive combination of skewness and Sharpe ratio.

The corresponding results for portfolios composed of the European index ESX and the ESX variance swaps are shown in Figure 5. These results are based on the shorter time period from 2000 to 2004. The main observations are very similar to the DAX analysis. In particular, there is once again a sharp distinction between skewness driven optimal portfolios and Sharpe ratio driven investments. This separation is possible even if the investor population represents a continuum of α - and β -preferences (see the jump of asset weights in Figure 5c). If we incorporate kurtosis preference into the optimization, investors with a strong preference for low kurtosis will select the better of the two $KT^{\min}$ -portfolios. But once again, the transition from the $SK^{\max}$ - to the $KT^{\min}$ -portfolio is very abrupt which means that the mixture of both portfolios is not attractive.

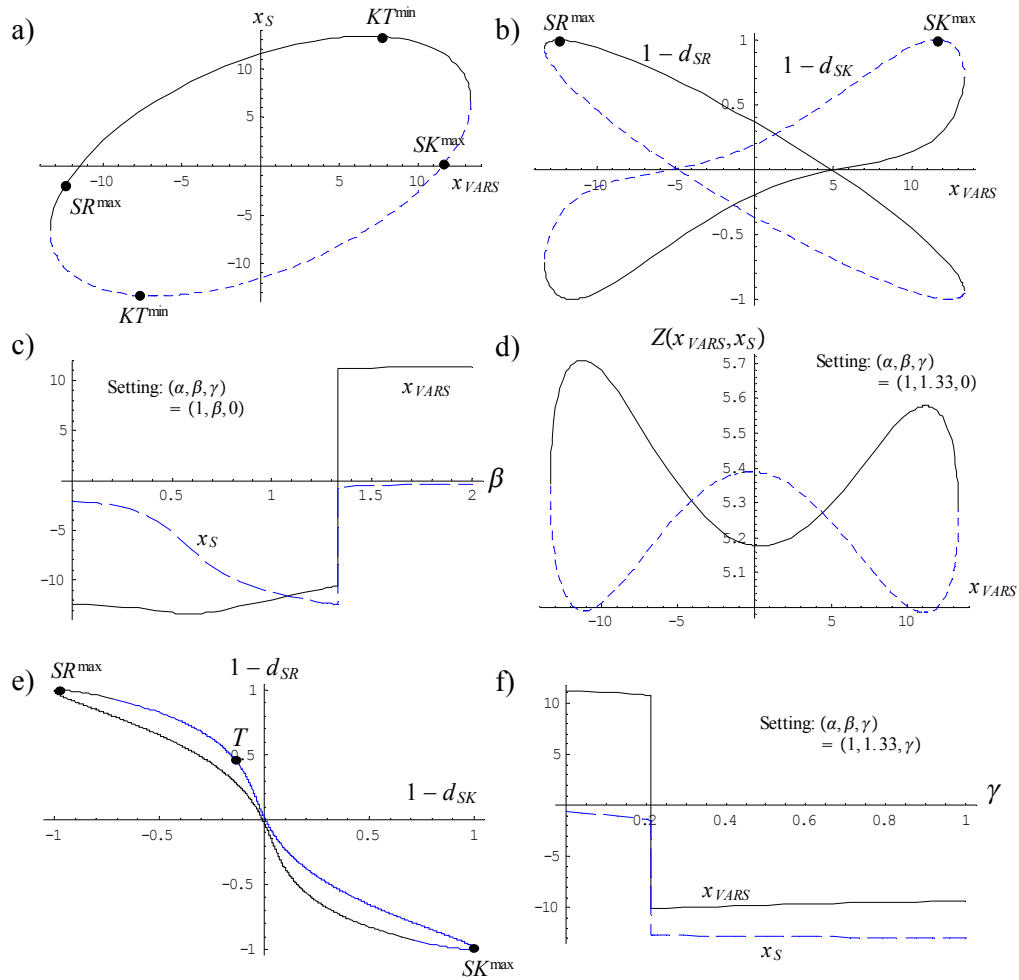


Figure 4: PGP optimization for portfolios of DAX, DAX variance swaps and the risk-free asset based on historical returns over the time period from 1995 to 2004. The lines show the coordinates of all unit variance portfolios. In figures a), b) and d), the solid lines always represent the same set of unit variance portfolios, whereas the coordinates of the remaining portfolios are represented by broken lines.

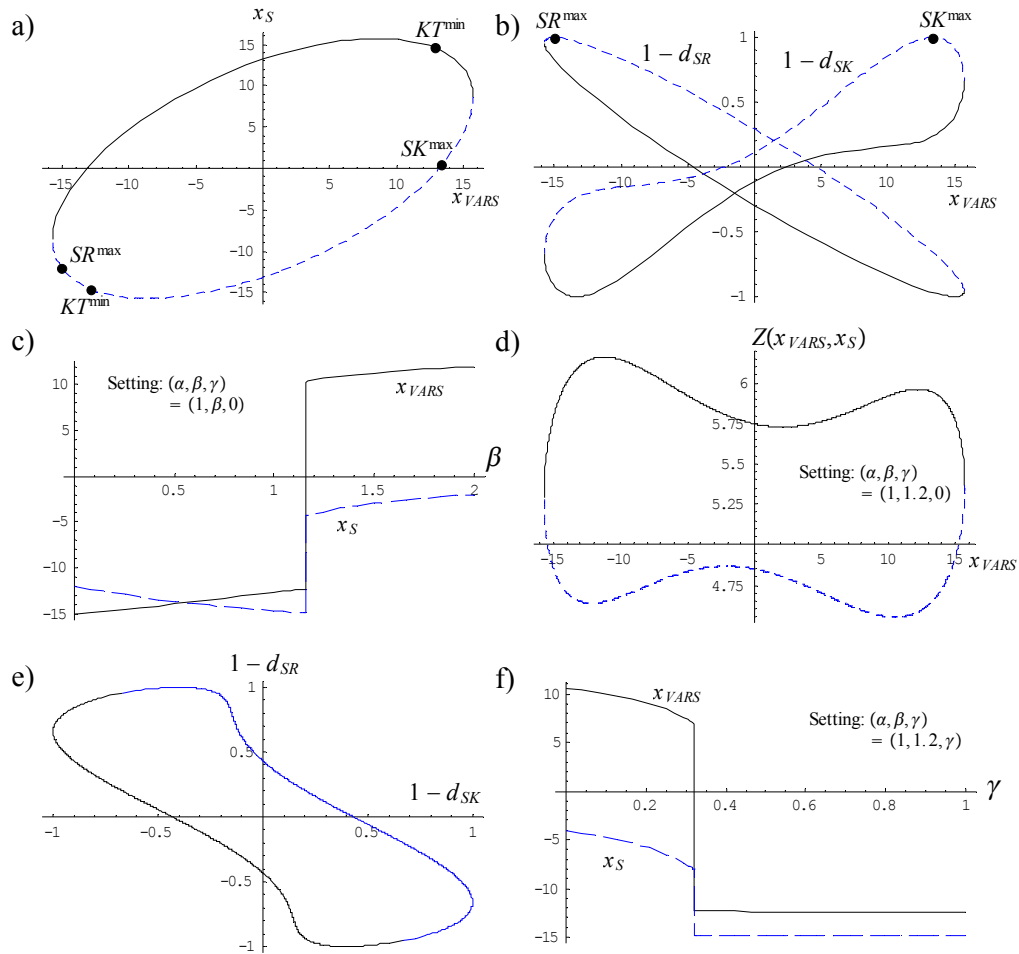


Figure 5: PGP optimization for portfolios of ESX, ESX variance swaps and the risk-free asset based on historical returns over the time period from 1995 to 2004. The lines show the coordinates of all unit variance portfolios. In figures a), b) and d), the solid lines always represent the same set of unit variance portfolios, whereas the coordinates of the remaining portfolios are represented by broken lines.

4 Conclusion

Mean returns of DAX and ESX variance swaps for the time period from 1995 to 2004 were strongly negative. This is compatible with the observation of a negative volatility risk premium reported in previous studies for the US and Europe. Part of the negative premium can be attributed to the negative correlation of variance swap returns with stock market index returns. However, beta alone does not suffice to explain its magnitude (see Carr and Wu (2005), Hafner and Wallmeier (2007)). As a consequence, mean-variance efficient portfolios are characterized by sizable short positions in variance swaps. Typically, the stock index is also sold short to make use of the negative return correlation with variance swaps and in that way achieve a better portfolio diversification. These results are robust with respect to reasonable estimation errors for expected returns. Results are also similar if we take the perspective of an investor who maximizes expected power utility.

Because expected utility maximization is based on the entire return distribution, it implicitly takes higher moments such as skewness and kurtosis into account. It is not surprising, however, that just one risk aversion parameter cannot capture the investors' heterogeneous preferences with respect to the higher moments of the portfolio return distribution. To be able to explicitly control such preferences, we use a variant of the Polynomial Goal Programming method (PGP) proposed by Lai (1991). This method attempts to find an investor-specific optimal combination of the degrees of achievement of different objectives. In our analysis we assume that investors strive for a high Sharpe ratio, high skewness and low kurtosis. This is in line with standard assumptions on economic behavior of investors. Unfortunately, the individual objectives are highly conflicting. In particular, the short positions required to profit from negative variance swap returns inevitably lead to an undesired negative skewness. Our analysis shows that a balanced trade-off between Sharpe ratio and skewness often does not exist. Investors tend to the extreme portfolios (Sharpe ratio driven, skewness driven or kurtosis driven) and avoid to be stuck in the middle. This 'all-or-nothing' characteristic is reflected in jumps of asset weights when certain thresholds of preference parameters are crossed.

In this study, we did not investigate whether the variance risk premium can be regarded as an adequate compensation for negative skewness and excess kurtosis. To answer this question, an equilibrium model incorporating higher moments of return distributions is needed. But the analysis clearly shows to what degree a short-selling strategy implies a negatively skewed distribution. It is therefore important to implement effective risk management procedures.

An important question for practitioners is whether the negative risk premium for realized variance is a temporary phenomenon or results from a structural imbalance that is likely to persist in the future. On the one hand, more and more hedge funds are willing to exploit the risk premium and increasingly act as variance sellers. On the other hand, the vast majority of market participants, among them most large institutional investors, are still reluctant to sell variance because it implies selling large amounts of out-of-the-money put options. This view is supported by the stability of the smile structure during recent years. Thus, there is some reason to believe that our analysis of historical returns is indicative of future return distributions and relevant for current investor decisions.

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